

Privacy-Preserving Recommendation Systems

Mid-Term Project Report

Mainak Sarkar (230619)

Shrasti Dwivedi (230982)

Aditya Anand (230065)

Shravan Agrawal (230984)

CS670, IIT Kanpur • 12 September 2026

Project status

OblivRec composes the two systems the instructor pointed us to: **NUDGE**'s three-server private matrix factorisation with **PIRSONA**'s DPF-based private retrieval. That composition is the contribution. **NUDGE**'s own non-goals state that it *“relies on other means . . . to let users fetch data items in a private way”*, and **PIRSONA**'s training core is superseded — so each paper leaves open exactly what the other supplies, and nobody has connected them.

Since the survey we have built and verified the **serving and delivery half (S1)** end to end on MovieLens-100K, against a model factorised in the clear. `demo --user 42 --k 10` splits a user's embedding across three servers, scores all 1682 items locally on each, lets the user reconstruct and select the top ten, and then fetches each recommended record by two-server DPF-PIR — returning ten real film titles, each verified byte-exact against a cleartext lookup. Private *training* (S2) begins now and is due for the final report; no claim is made about it here.

The strongest result is a measurement neither base paper reports. Because **NUDGE** publishes the item embedding matrix in the clear by design, an observed fetch is not one bit of information but a projection onto a known basis. We show that **ten observed fetches recover a user's private embedding to $\cos = 0.84$ and predict 56% of their next twenty recommendations** — which is the quantitative reason the delivery layer has to be private at all.

Per-member contributions

Mainak Sarkar (230619) — W1, FSS core and delivery

Wrote the entire implementation to date; every commit in the repository is evidence for this section.

- **The distributed point function**, from the Boyle–Gilboa–Ishai construction rather than imported. Verified *exhaustively*: every α against every x to 16 domain bits on $b = 64$ and 15 on $b = 128$ — 7.16×10^9 leaf checks in 27 minutes. Keys are 227 B at the demo's operating point against 16 KB for a naive query.
- **Found and fixed a real security bug** in key generation: it seeded a Mersenne Twister and used the output as DPF seed material. The seeds *are* the secret and MT19937 is reconstructible from 624 outputs, so an adversary with enough key material could rebuild the tree and recover α . Replaced with the OS CSPRNG, abort-on-failure.

- **The catalogue and the DPF-PIR read layer** — fixed-width 256 B records, and a server answer that is one EvalFull plus an unconditional inner product, with no data-dependent branch and no data pattern, since every coefficient is a pseudorandom share.
- **The serving path** — 2-of-3 replicated sharing and $\mathbf{a} \cdot \mathbf{B}$ computed locally per server, which needs no multiplication protocol because $\mathbf{B}$ is public. All 1682 scores match the independent Python oracle *bit for bit*.
- **The leakage analysis** described above, with a popularity control so the result cannot be an artefact of popular items, plus the **B1/B5 baselines** pricing the private read honestly (446× fewer bytes than downloading the catalogue, but slower in CPU than both baselines; crossover at 8.0 Gbit/s), and docs/threat-model.md.
- **Four real bugs found by tests rather than by inspection**, including a Python/C++ rounding-mode disagreement and a quantise-then-multiply error that shifted 1650 of 1682 reference scores by an amount any tolerance-based check would have accepted.
- All 11 sections of the survey report, the bibliography (66 DBLP-verified entries), and the 17-page technical report offered as an appendix.

Shrasti Dwivedi (230982) — W3, factorisation and serving

Milestone 1: delivered Part II of the recorded presentation (slides 10–16).

Implementation contributions for this milestone are to be supplied by Shrasti and are not recorded in the repository at the time of writing. This section is deliberately left factual rather than filled in on her behalf.

Aditya Anand (230065) — W2, 3PC substrate

Contributions for this milestone are to be supplied by Aditya and are not recorded in the repository at the time of writing.

Shravan Agrawal (230984) — W4, evaluation and security

Contributions for this milestone are to be supplied by Shravan and are not recorded in the repository at the time of writing.

Next steps before 6 November

Private training (S2) is the remaining half and starts now: secret-shared power iteration, with Trunc_{*t*} and ApproxNormalize as the only interactive steps. Two items were deferred from this phase and are named rather than dropped — the three-process networking, so that message-size obliviousness is *demonstrated* on a wire and not only argued from the construction, and the wide-area measurements that need it. The schedule lives in PHASES.md.